Week 2, Ridge regression

CAPP 30254, Machine Learning for Public Policy

Reminder: Linear machine learning model

Recall that a linear machine learning model makes predictions:

$$\widehat{y} = \mathbf{w}^T \mathbf{x} = w_1 x_1 + w_2 x_2 + \ldots + w_p x_p + \text{bias}$$

for a weight vector $\mathbf{w}$. The weight vector $\mathbf{w}$ is found by minimizing a loss function $\ell(\mathbf{w})$.

Recall that the least squares loss function $\ell(\mathbf{w})$ is:

$$\ell(\mathbf{w}) = \sum_{i=1}^{n} (y_i - \underbrace{\widehat{y}_i}_{= \underline{w}^T \underline{x}_i})^2$$

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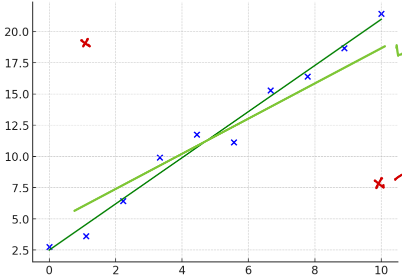
Least squares is sensitive to outliers

LS prone to overfitting:
1. Fits noise and outliers
2. Correlated features lead to big weights

Why? Large errors grow exponentially. So, outliers can dominate the total loss and have a disproportionately large impact on the model.

$$l_i(\underline{w}) = \underbrace{(y_i - \hat{y}_i)}_{\text{big error}}{}^{2}$$



LS with outliers

outliers

[1]

[1] Generated by ChatGPT with the prompt "can you draw a least squares fitted line with 10 data points?"

Regularization

Large values in the weight vector **w** can mean a model is complex. (overfitting)

Regularization is a technique that can help prevent overfitting and control the complexity of a model. It works by adding a *penalty term* to the loss function of a model.

Some popular penalty terms are:

$L_p = \left(\sum_i |w_i|^p\right)^{1/p}$

- L1: $\sum_i |w_i|$
- L2: $\sum_i w_i^2$
- Elastic net: combination of L1 and L2

Ridge regression

Ridge regression adds an L2 penalty term to the least squares loss function:

least squares + L2 penalty

$$\ell(\mathbf{w}) = \sum_{i=1}^{n}(y_i - \widehat{y}_i)^2 + \lambda \sum_{i=1}^{n} w_i^2$$

new loss

small λ: small regularization

large λ: weights shrink a lot

can become underfit

The parameter $\lambda \geq 0$ is the *strength* of the penalty.

what if $\lambda < 0$?

The penalty term encourages weights to stay small, helping to prevent overfitting by discouraging large values in **w**.

minimize $\ell(\underline{w}) \rightarrow \sum_i w_i^2$ stays small

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Graphically: How does ridge regression work?

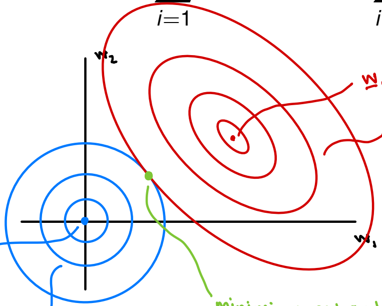
Recall that to find $\mathbf{w}_{RR}$ we minimize the loss the function ℓ:

red blue

$$\ell(\mathbf{w}) = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2 + \lambda \sum_{i=1}^{n} w_i^2$$

$\underline{w}_{RR} = \underset{\underline{w}}{\arg\min}\ \ell(\underline{w})$

$\underline{w} \in \mathbb{R}^2$



"contour lines" "level curves" "isosurface"

Mathematically: How does ridge regression work?

Like least squares, ridge regression has a close-formed solution:

$(U\Sigma V^T)^T = V\Sigma^T U^T$

$UU^T = I \;\downarrow\; U^T = U^{-1}$

$$\mathbf{w}_{RR} = (X^TX + \lambda I)^{-1}X^T\mathbf{y}$$

use the SVD!

$$\underline{w}_{RR} = \left((U\Sigma V^T)^T(U\Sigma V^T) + \lambda I\right)^{-1}(U\Sigma V^T)^T y$$

$$= (V\Sigma^T \underbrace{U^T U}_{=I} \Sigma V^T + \lambda \underbrace{I}_{= VV^T = VIV^T})^{-1}(V\Sigma^T U^T) y$$

$$= (V\Sigma^T\Sigma V^T + V\lambda I V^T)^{-1} V\Sigma U^T y$$

$$= (V(\Sigma^T\Sigma + \lambda I)V^T)^{-1} V\Sigma U^T y$$

$$= V(\Sigma^T\Sigma + \lambda I)^{-1} \underbrace{V^T V}_{=I} \Sigma U^T y$$

$\approx$ SVD

$$\Sigma^T\Sigma = \begin{bmatrix} \sigma_1^2 & & 0 \\ & \ddots & \\ 0 & & \sigma_p^2 \end{bmatrix}$$

DNE if any $\sigma_i = 0$

$$\Sigma^T\Sigma + \lambda I = \begin{bmatrix} \sigma_1^2+\lambda & & 0 \\ & \ddots & \\ 0 & & \sigma_p^2+\lambda \end{bmatrix}$$

always invertible $\lambda > 0$

$$\underline{w}_{RR} = V(\Sigma^T\Sigma + \lambda I)^{-1}\Sigma^T U^T y$$

is always computable!

what about overfitting?

$$(\Sigma^T\Sigma + \lambda I)^{-1}\Sigma^T = \begin{bmatrix} \frac{\sigma_1^2}{\sigma_1 + \lambda} & & 0 \\ & \ddots & \\ 0 & & \frac{\sigma_p^2}{\sigma_p + \lambda} \end{bmatrix}$$

can help when σ_i is very small:

if $\sigma_i \to 0$

$\frac{1}{\sigma_i} \to$ huge

$\frac{\sigma_i^2}{\sigma_i + \lambda} \to \frac{1}{\lambda}$

[2]

Ex. $x = 10^{-6}$

$\frac{1}{x} = 10^{6}$

[2] Generated by GPT-4, DALL-E 3 after the prompt: “Hi, could you please make a minimalistic logo for a machine learning class...”